Programming assignment #2 Gaussian Naive Bayes
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CS445 ML spring 2026

For this assignment, I had to implement a Gaussian Naive Bayes classifier to predict whether emails are spam or not. The data set is taken from the UCI ML repository. The data set contains over 4000 emails with 57 features for an email that are binary classified, 1 or 0.
I did what is called a stratified split of the data so that both halves contain roughly the same spam rate of about 40%. Which is the same as the full dataset.
For training, I needed to compute the prior probability of each class and the mean and std of the 57 features for each class. For any std of zero, I assigned a value of 0.0001 so I avoid dividing by 0.
To classify, I used the log sum version of the Naive Bayes for each email. For both classes, I added the log before the sum of the log Gaussian likelihoods across all 57 features. Then I can predict which class got the higher score. I used logs to avoid floating-point underflow

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  4601 emails, 57 features, spam rate = 0.394

Step 1: splitting into train/test sets
  training: 2301 emails (spam rate: 0.394)
  test:     2300 emails (spam rate: 0.394)

Step 2: training the model
  P(not-spam) = 0.6058
  P(spam) = 0.3942

Step 3: classifying 2300 test emails

Results:
  Accuracy  = 0.8239 (82.39%)
  Precision = 0.7072
  Recall    = 0.9437

Confusion matrix:

                       Predicted NOT-spam    Predicted SPAM
--------------------------------------------------------------
Actual NOT-spam               1040                 354
Actual SPAM                    51                  855
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That is what I got after I ran the program. This model correctly classified 1895 out of the 2300 test emails. It caught 855 out of the 906 actual spam emails. But it was also incorrectly classified 354 normal emails as spam. And only 51 actual spam emails were missed by the program. The program will rather over-flag something as spam than let a spam email get classified as normal.

Are the attributes independent?

I don't think that the attributes are independent of each other. Most spam emails would use certain words together. Words like "free" and "money" could go together. Some features may depend on one another. It would be very surprising if all 57 features were independent.

Does Naïve Bayes do well on this problem in spite of the independence assumption?

Yes, I think it does well with a 82.39% accuracy. The independence assumption isn't really true, but I think the model still works because more actual spam emails get classified as spam. We just need to get the ranking rights instead of probabilities.

Speculate on other reasons Naïve Bayes might do well or poorly on this problem?

A reason why this Naive Bayes does well is because of the clear signals of their own features. Like the word "free" will not pop up on normal emails as often as spam. All of the 57 features pointing in the same direction will add up, and it will be safe to classify some spam or normal emails.

A reason it might do poorly is that most of the words would be classified as 0. Emails might not contain any tracked words. Therefore, the Gaussian formula won't work well because there's a lot of 0's